EXPERIMENT 6 (HARIN T_192412412)

1. Aim

To simulate and analyze the frequency-conversion characteristics of a microwave mixer using MATLAB and determine the intermediate frequency, sum frequency, and conversion loss.

2. Objectives

1. To study microwave mixer frequency conversion.
2. To identify the difference and sum frequency components.
3. To calculate conversion loss and evaluate mixer performance.

3. Apparatus Required

1. Computer or laptop
2. MATLAB software

4. Procedure

1. Open MATLAB and create a new script.
2. Define the RF frequency, LO frequency, and input power levels.
3. Generate the simulated mixer output spectrum.
4. Identify the difference and sum frequency components.
5. Determine the IF output power and conversion loss.
6. Plot the output spectrum and observe the frequency components.

5. MATLAB Program

```
clc; clear; close all;

fRF=10; fLO=9.5; PRF=-10; PLO=7;

fIF=abs(fRF-fLO); fsum=fRF+fLO;

PIF=-18; CL=PRF-PIF;

fprintf('Suraj Narasimhan Potty | 192412442\n');

fprintf('IF Frequency = %.2f GHz\n',fIF);

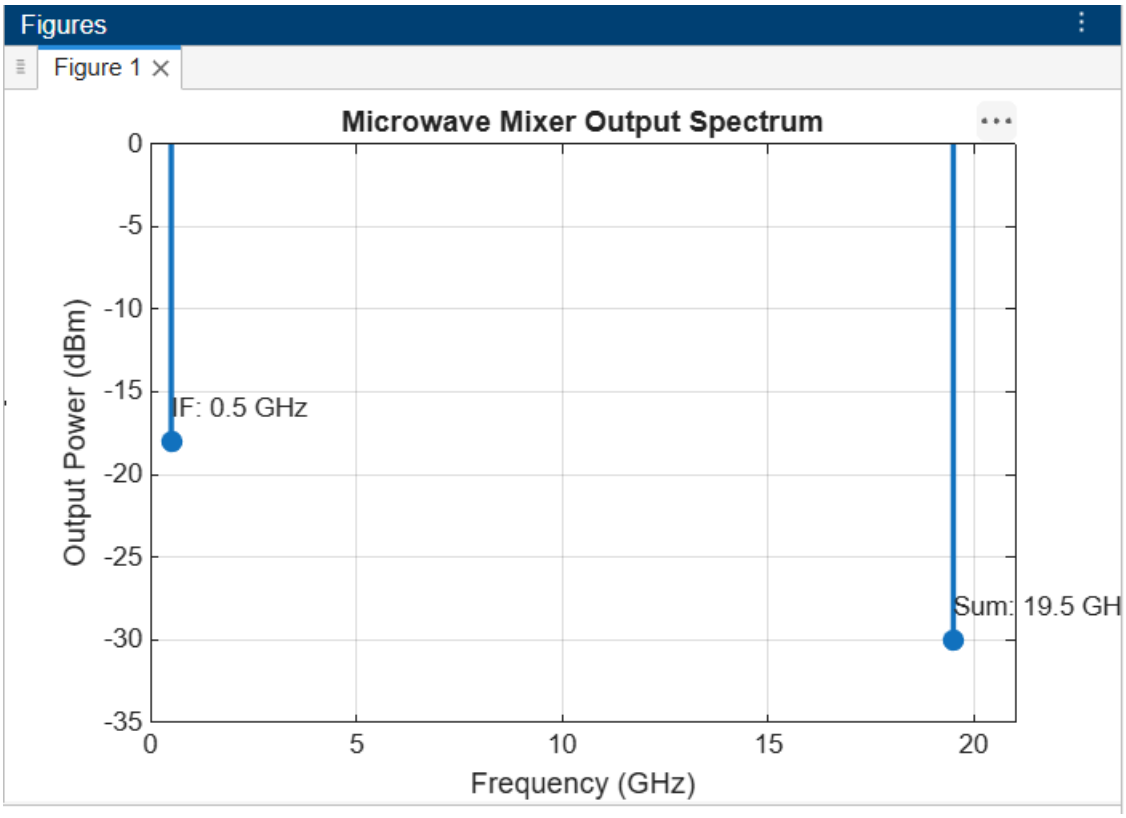
fprintf('Sum Frequency = %.2f GHz\n',fsum);

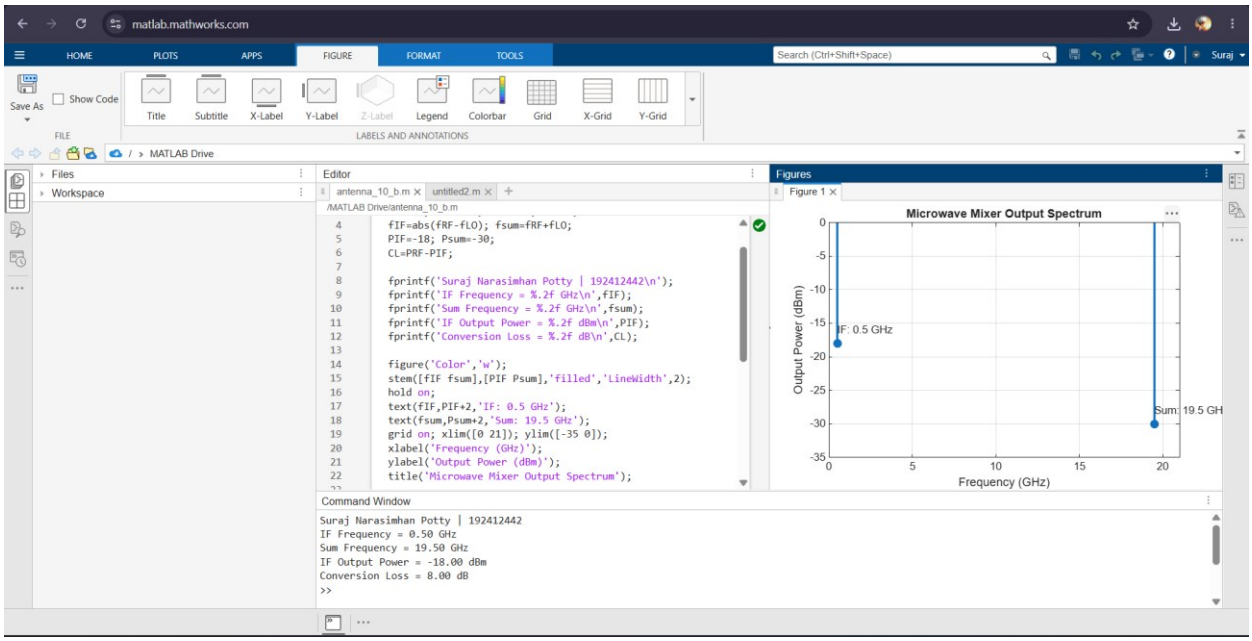
fprintf('IF Output Power = %.2f dBm\n',PIF);

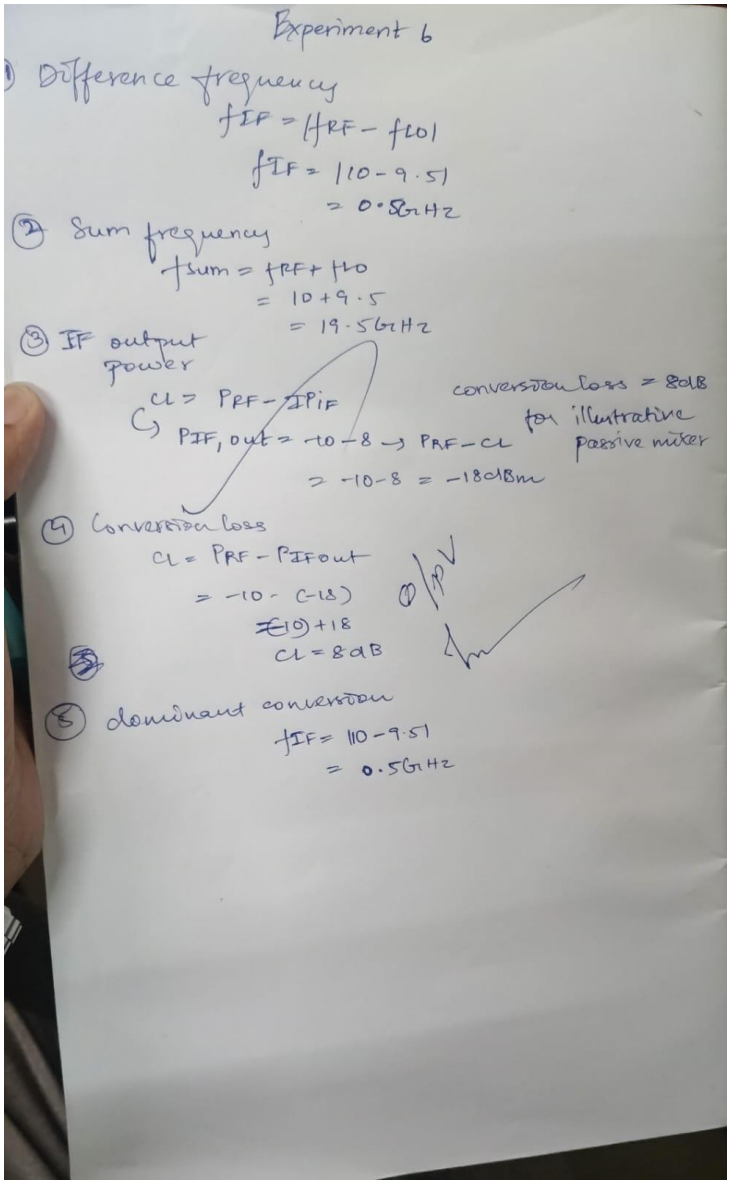
fprintf('Conversion Loss = %.2f dB\n',CL);
```

```
F=[fIF fsum]; P=[PIF -30];
figure('Color','w');
stem(F,P,'filled','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Output Power (dBm)');
title('Microwave Mixer Output Spectrum');
xlim([0 21]);
```

OUTPUT:




matlab.mathworks.com
HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Suraj
Save As
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
/ > MATLAB Drive
Files
Workspace
Editor
antenna_10_b.m
untitled2.m
/MATLAB Drive/antenna_10_b.m
4 fIF=abs(fRF-fLO); fsum=fRF+fLO;
5 PIF=-18; Psum=-30;
6 CL=PRF-PIF;
7
8 fprintf('Suraj Narasimhan Potty | 192412442\n');
9 fprintf('IF Frequency = %.2f GHz\n',fIF);
10 fprintf('Sum Frequency = %.2f GHz\n',fsum);
11 fprintf('IF Output Power = %.2f dBm\n',PIF);
12 fprintf('Conversion Loss = %.2f dB\n',CL);
13
14 figure('Color','w');
15 stem([fIF fsum],[PIF Psum],'filled','LineWidth',2);
16 hold on;
17 text(fIF,PIF+2,'IF: 0.5 GHz');
18 text(fsum,Psum+2,'Sum: 19.5 GHz');
19 grid on; xlim([0 21]); ylim([-35 0]);
20 xlabel('Frequency (GHz)');
21 ylabel('Output Power (dBm)');
22 title('Microwave Mixer Output Spectrum');
Figures
Figure 1
Microwave Mixer Output Spectrum
Output Power (dBm)
Frequency (GHz)
IF: 0.5 GHz
Sum: 19.5 GH
0
-5
-10
-15
-20
-25
-30
-35
0
5
10
15
20
Command Window
Suraj Narasimhan Potty | 192412442
IF Frequency = 0.50 GHz
Sum Frequency = 19.50 GHz
IF Output Power = -18.00 dBm
Conversion Loss = 8.00 dB
>>


Experiment 6
1) Difference frequency
fIF = |fRF - fLO|
fIF = |10 - 9.5|
= 0.5GHz
2) Sum frequency
fsum = fRF + fLO
= 10 + 9.5
= 19.5GHz
3) IF output power
CL = PRF - PIF
PIF, out = -10 - 8 → PRF - CL
= -10 - 8 = -18dBm
conversion loss = 8dB
for illustrative
passive mixer
4) Conversion loss
CL = PRF - PIFout
= -10 - (-18)
= 10 + 18
CL = 8dB
5) dominant conversion
fIF = 110 - 9.51
= 0.5GHz

6. Reflection

I learned how a microwave mixer converts RF and LO signals into sum and difference frequency components. I also understood how to determine the intermediate frequency and conversion loss and visualize the output spectrum using MATLAB.

7. Result

The frequency-conversion characteristics of a microwave mixer were studied using MATLAB. The theoretical IF and sum frequencies were determined as 0.5 GHz and 19.5 GHz, respectively. Using an assumed IF output power of −18 dBm, the conversion loss was calculated as 8 dB.